Cem Gokmen

CS PhD Candidate at Stanford, Specializing in Embodied AI and Robotics Stanford, CA 94305
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Research Statement

My research focuses on developing machine learning algorithms that enable robots to perform complex, long-horizon tasks in real-world environments. I specialize in real-to-sim and sim-to-real transfers, imitation learning, reinforcement learning, and leveraging foundation models (LLMs/VLMs), with extensive experience building high-fidelity simulation environments and benchmarks to accelerate research in embodied AI.

Seeking a full-time research role starting June 2026 to further drive exciting robotics & embodied AI research.

Education

Stanford University, Stanford, CA Sep. 2022 - Expected Jun. 2026
Ph.D. in Computer Science, **Advisor:** Prof. Fei-Fei Li

Stanford University, Stanford, CA Sep. 2020 - Jun. 2022
M.Sc. in Computer Science GPA: 4.03

Georgia Institute of Technology, Atlanta, GA Aug. 2016 - Dec. 2018
B.Sc. in Computer Science with Undergraduate Research Certification GPA: 3.83

Work Experience

Research Scientist Intern, FAIR Robotics, Meta
Host: Roozbeh Mottaghi. Menlo Park, CA. Jun. 2025 - Present

- Developed a method to generate physically stable, object-decomposed robot simulation scenes from house tour / real estate videos.

Research Scientist Intern, NVIDIA Research
Hosts: Jim Fan, Yuke Zhu. AI Algorithms Group, Santa Clara, CA. Jun. 2023 - Sep. 2023

- Leveraged LLMs to generate diverse demonstrations for learning end-to-end transformer policies for long-horizon robotics tasks.

AI Resident, Google X / Everyday Robots
Hosts: Mohi Khansari, Daniel Ho. Mountain View, CA. Jun. 2022 - Sep. 2022

- Enabled robots to predict their success probability from imitation learning policy rollouts, allowing them to proactively ask for human help when uncertain.

Software Engineer II, Google
YouTube Premium Team, San Bruno, CA Feb. 2019 - Sep. 2020

- Led full-stack feature development (Python/C++ backend, Android/iOS/Web) for YouTube Premium user journeys.

Highlighted Publications

C. Li*, R. Zhang*, J. Wong*, **C. Gokmen***, S. Srivastava*, R. Martin-Martin*, C. Wang*, G. Levine*, *et al.*, "BEHAVIOR-1K: A benchmark for embodied AI with 1,000 everyday activities and realistic simulation," in *6th Annual Conference on Robot Learning*, 2022, **(Co-First Author, CoRL Oral, Best Paper Nominee)**

- **Currently hosting the 1st BEHAVIOR Challenge at NeurIPS 2025.** We evaluate robots on 50 full-length household tasks in the realistic BEHAVIOR-1K environment. We also released 10,000 teleoperated expert demonstrations (1200+ hours), making it the biggest long-horizon task demonstration dataset available.
- I co-develop and maintain **OmniGibson** and the older-generation **iGibson**, the high-fidelity simulation environments to simulate the tasks, built on Omniverse Isaac Sim and PyBullet, respectively. Visit https://github.com/StanfordVL/BEHAVIOR-1K for more info.
- I developed the process and managed the 10+ person team that acquired and created the 6k-object, 50-scene BEHAVIOR-1K dataset, a massive undertaking involving full cross-integration of the assets with the 1,000 task definitions and 4,000 nouns in the BEHAVIOR-1K knowledgebase. Visit https://behavior.stanford.edu/knowledgebase/ for some examples.

C. Gokmen, D. Ho, and M. Khansari, "Asking for help: Failure prediction in behavioral cloning through value approximation," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, 2023

- Introduced **Behavioral Cloning Value Approximation (BCVA)**, a novel method for imitation learning policies to predict failures and proactively request help.
- Deployed on real-world robots at Google, achieving 86% precision in help requests over 2000+ trials on complex manipulation tasks.

M. Li, S. Zhao, Q. Wang, K. Wang, Y. Zhou, S. Srivastava, **C. Gokmen**, T. Lee, *et al.*, "Embodied agent interface: Benchmarking LLMs for embodied decision making," in *Neural Information Processing Systems Datasets and Benchmarks Track*, 2024, **(NeurIPS Benchmark Track Oral)**

- Developed the **Embodied Agent Interface**, a standardized framework for rigorously benchmarking LLMs on embodied decision-making tasks.
- This framework allows standardized and fine-grained assessment of LLMs' abilities in handling complex, embodied decision-making scenarios, pinpointing issues like hallucinations and planning errors across 15 LLMs evaluated on benchmarks such as BEHAVIOR and VirtualHome.

Y. Ge*, Y. Tang*, J. Xu*, **C. Gokmen***, C. Li, W. Ai, B. J. Martinez, A. Aydin, *et al.*, "Behavior vision suite: Customizable dataset generation via simulation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2024, **(Co-First Author, CVPR Highlight)**

- Created the **BEHAVIOR Vision Suite (BVS)**, a toolset for generating customizable synthetic data to benchmark computer vision models.
- Enables controlled experiments on model robustness, scene understanding, and sim-to-real transfer by manipulating scene, object, and camera parameters, bypassing limitations of real-world datasets like poor labeling and diversity.

Other Publications

Y. Jiang, R. Zhang, J. Wong, C. Wang, Y. Ze, H. Yin, **C. Gokmen**, S. Song, *et al.*, "Behavior robot suite: Streamlining real-world whole-body manipulation for everyday household activities," in *9th Annual Conference on Robot Learning*, 2025

T. Dai, J. Wong, Y. Jiang, C. Wang, **C. Gokmen**, R. Zhang, J. Wu, and L. Fei-Fei, "ACDC: Automated creation of digital cousins for robust policy learning," in *8th Annual Conference on Robot Learning*, 2024

C. Gokmen, R. Zhang, S. Srivastava, C. Li, M. Lingelbach, R. Martin-Martin, S. Savarese, J. Wu, *et al.*, "Eye-BEHAVIOR: An eye-tracking dataset for everyday household activities in virtual, interactive, and ecological environments," *Journal of Vision*, Dec. 2022

S. Srivastava, C. Li, M. Lingelbach, R. Martín-Martín, F. Xia, K. E. Vainio, Z. Lian, **C. Gokmen**, *et al.*, "Behavior: Benchmark for everyday household activities in virtual, interactive, and ecological environments," in *Proceedings of the 5th Conference on Robot Learning*, 2022. [Online]. Available: `https://proceedings.mlr.press/v164/srivastava22a.html`

C. Li, F. Xia, R. Martín-Martín, M. Lingelbach, S. Srivastava, B. Shen, K. E. Vainio, **C. Gokmen**, *et al.*, "Igibson 2.0: Object-centric simulation for robot learning of everyday household tasks," in *Proceedings of the 5th Conference on Robot Learning*, 2022. [Online]. Available: `https://proceedings.mlr.press/v164/li22b.html`

Teaching

Head Teaching Assistant, CS 231N: Deep Learning for Computer Vision
Stanford University — Mar. 2024 - Jun. 2024

- Led team of 22 course assistants for Stanford's largest CS course (650+ students), planning assignments and exams, and mentoring student projects.
- See https://cs231n.stanford.edu/2024/reports.html for mentored projects.

Skills & Misc.

Languages: English (Fluent), Turkish (Native), French (Advanced), Spanish (Beginner).
Programming: Python, Java, C, C++, JavaScript, HTML, CSS.
Research: Reinforcement Learning, Imitation Learning, Physics Simulation, Robot Simulation, Motion Planning, Classical Robotics, Foundation Models (LLM/VLM/VLA)
Frameworks: PyTorch, TensorFlow, Ray, PyBullet, Omniverse, Isaac Sim, CUDA, Docker
Immigration Status: U.S. Permanent Resident / Green Card Holder